

Diego Penton

pentond27@gmail.com | linkedin.com/in/diegopenton | github.com/diegopenton | Lakeland, FL (Open to relocate)

EDUCATION

Florida Polytechnic University

Graduating: May 2027

Bachelor of Science (BS): Computer Science - Artificial Intelligence

Lakeland, FL

Relevant Coursework: Artificial Intelligence, Applied Deep Learning, Machine Learning, Natural Language Processing, Algorithm Design & Analysis, Data Structures, Secure Software Engineering, Operating Systems Concepts, Database I, Computational Linear Algebra, Statistics I, Investments, Financial Modeling & Analytics, Computer Networks, Object-Oriented Programming

WORK EXPERIENCE

Vulkan USA - IT Intern

May 2026 - Aug 2026

- Developed **AI and automation solutions** for U.S. manufacturing and documentation workflows, including **OCR tools** that converted scanned factory records into searchable digital text
- Built and evaluated a TrOCR digitization workflow with preprocessing and review steps, achieving 92% character accuracy and 72.28% exact serial accuracy on factory records**
- Worked with **SAP HANA** data workflows to understand enterprise storage and integration needs for future **AI-agent search**, indexing, and internal documentation systems

Florida Polytechnic University, Lakeland, FL - Admissions Ambassador

Aug 2024 - Present

- Lead **campus tours and information sessions** for prospective students and families, explaining academic programs, admissions requirements, student life, and campus resources
- Supported **1,500+ prospective students** with individualized guidance, answering admissions questions and connecting visitors with relevant academic and university resources
- Represent Florida Poly at campus visits, **recruitment events**, and information sessions, helping students and families navigate the university and admissions experience

University of Zurich, Zurich, Switzerland - Microscopy Intern

July 2022 - Aug 2022

- Created and standardized training guides covering **TEM, SEM, Cryo-EM**, and scientific image-analysis tools used in the department's microscopy research workflows
- Gained **hands-on experience** with **microscopy equipment and scientific imaging software**, building practical familiarity with imaging, analysis, and research-lab workflows
- Updated technical documentation and reorganized the department's **training database**, making microscopy procedures, software guidance, and research materials easier to access

PROJECTS

Project Argos - Multi-Agent AI Trading Research System | Python, LLMs, RAG, Docker, GitHub

June 2026 - Present

- Architecting a **multi-agent AI system** that converts financial research into structured, **testable trading strategies**
- Integrated local **LLM inference, RAG, persistent memory**, and specialized agents on **NVIDIA DGX Spark** using Docker
- Built an auditable corpus pipeline with provenance, rights classification, and **15,000+ indexed research records**

OCR Documentation Digitization Pipeline | Python, Hugging Face, PyTorch, TrOCR

May 2026 - Aug 2026

- Developed a **PyTorch/TrOCR pipeline** to convert scanned manufacturing tables into structured, searchable text
- Built preprocessing, labeling, and review workflows reaching **92% character** and **72.28% exact serial accuracy**
- Structured extracted records and validation outputs for search, indexing, and future **AI-agent integration**

AI Coding Assistant Comparative Analysis Framework | Python, APIs, Bandit

March 2026 - May 2026

- Designed **standardized prompts and API workflows** to compare AI coding assistants on repeatable Python tasks
- Evaluated code for correctness, readability, runtime performance, and security using **Bandit** and scoring criteria
- Organized generated code and benchmark results into a **repeatable workflow** for side-by-side model comparison

Graph-Based Airport Connectivity & Flight Route Optimization System | C++

April 2025 - May 2025

- Modeled airports and routes as **weighted graphs** to analyze connectivity, alternate paths, and route relationships
- Implemented **shortest-path, minimum spanning tree, and state-search algorithms** in C++ for route optimization
- Applied **graph search techniques** to identify reachable routes and connectivity patterns across airport networks

LEADERSHIP

Vice President of Marketing, Society of Hispanic Professional Engineers (SHPE)

May 2026 - Present

- Lead chapter marketing for **professional development**, recruitment, convention initiatives, and **sponsor outreach**
- Coordinate digital campaigns and promotional materials with a **seven-member board** and **regional partners**
- Increased monthly Instagram views by **~30K (150%)** and follower count by **35.5%** since becoming VP of Marketing

SKILLS

Programming: Python, C++, C, SQL

AI / Machine Learning: PyTorch, Hugging Face Transformers, TrOCR, LLMs, RAG, Multi-Agent Systems, OCR

Data & Quantitative: NumPy, Pandas, Matplotlib, yfinance, Backtesting, Walk-Forward Validation, GARCH

Tools & Platforms: Git, GitHub, Docker, Linux, vLLM, REST APIs, SAP HANA, NVIDIA DGX Spark, LaTeX